

# Zakhar Pashkin

## Senior ML Engineer | Computer Vision, Document AI & Agentic Systems

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[zack-dev-cm.github.io](https://zack-dev-cm.github.io) | [GitHub](#) | [LinkedIn](#)

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Senior ML Engineer with 8+ years developing computer vision and applied ML from research through deployment. Shipped food-recognition and OCR workflows at Carb Manager and built financial-document recognition at CFT. Delivered skin-analysis models, multimodal retrieval and a published model-profiling SDK. Currently developing document AI and 3D geometry workflows.

### EXPERIENCE

#### Senior ML Engineer

Riverstart | Jul 2026 - Present

R&D ML | Remote

- Develop and evaluate an on-premises document assistant for specialist case review, separating deterministic checks from model explanations. Build construction-plan analysis with structured quantities; link both workflows to source pages.
- Research computer vision for engineering-drawing interpretation and 3D geometry analysis.

#### ML / Computer Vision Engineer

Dermaself · Agnitra · Multimodal Video Search | Jun 2024 - Jun 2026

Remote

- Developed Dermaself's skin-analysis pipeline from guided capture and facial regions through pore/wrinkle segmentation and mobile/API integration. Restored matching wrinkle outputs across cloud and GPU runtimes in regression checks.
- Developed multimodal video search combining speech, on-screen text and visual signals. Delivered Qdrant hybrid retrieval with BM25 and dense embeddings, supported by evaluation and runtime profiling.
- Built Agnitra, a Python SDK and CLI for model profiling and inference optimization, with runtime telemetry, model graph representations and benchmark workflows. Published the package on PyPI.

[Agnitra on PyPI](#) | [Video search](#) | [Dermaself](#)

#### Senior Computer Vision Engineer

Wombat Apps LLC / Carb Manager | Jun 2022 - Jun 2024

Remote

- Shipped food-recognition and nutrition-label workflows across cloud, iOS and Android for Carb Manager, a nutrition product serving millions of customers.
- Combined detection, OCR and table parsing to turn nutrition-label images into structured nutrition data. Owned data preparation, model evaluation, API integration, mobile export and runtime profiling with mobile, backend and product teams.

## EARLIER EXPERIENCE

### Computer Vision Engineer

Center of Financial Technologies (CFT) | Jun 2019 - Jun 2022

Financial technology

- Built core detection, segmentation and OCR models for CFT's financial-document recognition platform using PyTorch and OpenMMLab. Developed active-learning and assisted-annotation workflows to reduce manual labeling.
- Optimized meter-recognition models for mobile use through architecture changes, quantization and runtime profiling. Mentored engineers on data quality and evaluation.

### Machine Learning Specialist & Project Mentor

School of AI | Oct 2018 - Oct 2019

Part-time community program

- Built ML learning software, coordinated remote contributors and mentored applied CV projects from dataset definition through evaluation and working demos.

## SELECTED PROJECTS & RESEARCH

**Neuralink datarepo** | Independent open-source contributions | 3 open PRs

Submitted fixes for ClickHouse null filtering, a runnable local quick start and Python wheel compatibility.

Published patches and real-backend regression evidence; awaiting upstream review as of 8 Sep 2026.

**InQuest** | Project-aware document agents

Built document agents with project-specific source routing and output storage using OpenAI Agents SDK and MCP. Added evaluation scenarios for retrieval selection and saving behavior.

**Calorio** | Maintained service

Built and maintain a Telegram AI nutrition service for meal logging from photos, voice and text, with a food diary and nutrition-goal tracking.

**LigninQC** | Research tooling | In progress

Built a reproducible literature-discovery and audit pipeline for computational lignin research, with publication-version grouping and traceable extraction schemas.

**Vehicle Lab** | Open-source engineering notebook

Built a Python and Three.js notebook for CAD inspection, design revisions and recorded simulation.

## TECHNICAL EXPERTISE

**Computer vision:** Python, PyTorch, OpenMMLab (MMEEngine, MMCV, MMDetection, MMSegmentation, MMOCR), OpenCV, YOLO, OCR, segmentation, CLIP/VLMs, ONNX Runtime, Core ML

**LLMs & agents:** LangGraph, LangChain, OpenAI Agents SDK, MCP, Pydantic, LlamaIndex, RAG, Qwen/vLLM, Qdrant, Neo4j, pgvector, tool calling, evaluation and human review

**ML systems:** FastAPI, PostgreSQL, Redis, Docker, Kubernetes, AWS/GCP, Cloud Run, GitLab CI/CD, GitHub Actions, MLflow/ClearML, Prometheus/Grafana

## EDUCATION

Siberian Federal University | Computer Science & Microelectronic Engineering | 2007-2012